Capital Composition without Standard Cointegration? A Unified Empirical Admissibility Audit of Chilean Heterogeneous Capital and External Pressure (1920–2024)

Chapter 2 Research Group
Department of Economics, UMass Amherst

July 2026

Abstract

This paper asks whether Chile's nonresidential structures (K^{NR}) form a stable long-run accommodating envelope for machinery and equipment (K^{ME}, 1920–2024), and whether World Historical Balance of Payments (WBOP) pressure conditions that relationship. We reproduce the data-first sequence used in the United States capital-composition audit while enforcing the current integration-order, break, bootstrap, and notation rules. Four results determine the verdict. First, machinery and structures rise together over the century, but the normalized relation $k_t^{\mathrm{NR}} = \alpha + \beta_{\mathrm{ME}} k_t^{\mathrm{ME}} + \varepsilon_t$ is unstable: Bai–Perron dates 1934, 1949, 1964, and 1994, while CUSUM rejects parameter constancy. Second, the full century, Classic ISI, and post-1990 windows classify both capital stocks as $I(2)$-risk; the 1920–1938 and 1974–1989 crisis windows remain unresolved. No window supplies the common $I(1)$ prerequisite for Johansen, IM–OLS, FM–OLS, or DOLS, so the paper reports no pseudo-cointegrating coefficient or adjustment speed. Third, manufactured imports, the current account, and net foreign income pass as $I(1)$, while gross foreign liabilities pass as $I(0)$; their model roles are identified, but the failed capital prerequisite blocks long-run WBOP conditioning. Fourth, annual WBOP financing compressions in 1937, 1976, 1999, and 2023 coincide with a mean post-event machinery-minus-structures accumulation gap of -0.57 percentage points, yet the 999-draw interval $[-2.36, 0.55]$ includes zero. Chile's external accounts remain relevant to the realization of mechanization, but the current capital stocks do not identify a standard long-run choice-of-technique vector.

1 Introduction: Choice of Technique and Chilean Capital Heterogeneity

Capital accumulation combines physically distinct commitments. Machinery and equipment (K^{ME}) records the intensive choice of technique: the equipment, imported complements, and production systems through which firms reorganize labor and raise technical productivity. Nonresidential structures (K^{NR}) record the extensive envelope: factories, warehouses, transport installations, and other durable plant required to house and connect that machinery. In Chile, machinery is maintained as the more import-dependent margin and structures as the predominantly domestic margin. This is an analytical restriction whose observable implications must be tested; the capital accounts do not directly measure import content.

The Layer-2 relation is normalized with structures on the left:

$$k_t^{\mathrm{NR}} = \alpha + \beta_{\mathrm{ME}} k_t^{\mathrm{ME}} + \varepsilon_t. \tag{1}$$

The interpretation follows the normalization. A coefficient $\beta_{\mathrm{ME}} < 1$ means machinery expands faster than its structural envelope; $\beta_{\mathrm{ME}} > 1$ means structures respond more than proportionally to machinery. Exact proportionality requires $\beta_{\mathrm{ME}} = 1$, and stationarity of

$$\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NR}} \tag{2}$$

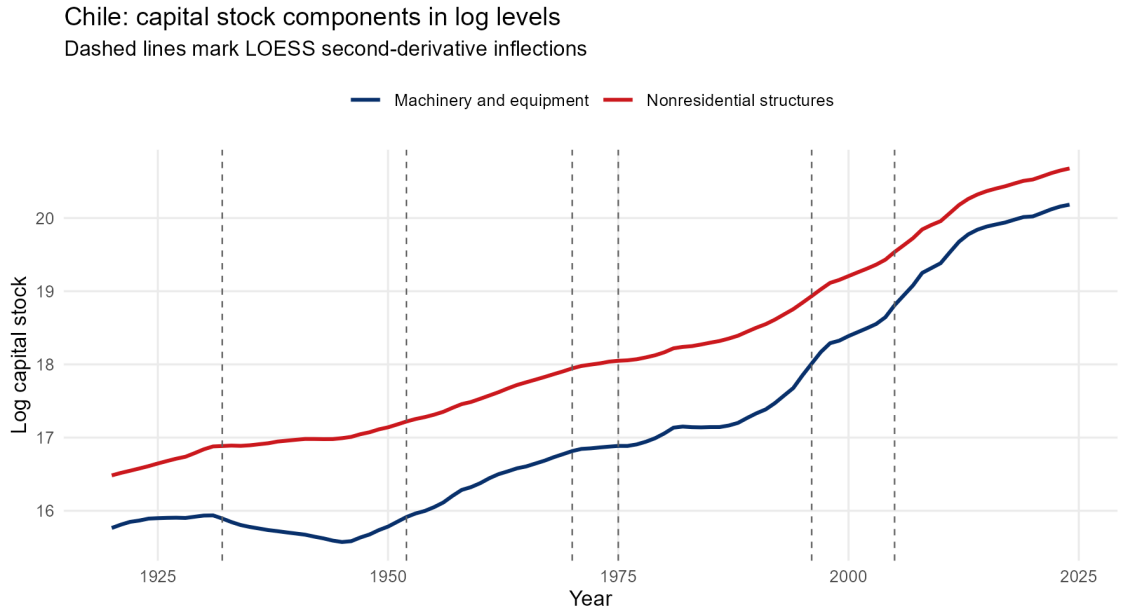
requires testing that exact restriction within a valid cointegrating system. A coefficient merely near one does not establish it.

The empirical question precedes the estimator. Standard Johansen and modified-OLS inference requires the two capital stocks to share the appropriate integration order. Chile's century-long stocks contain strong stochastic persistence and repeated structural change; imposing $I(1)$ because both series trend upward would manufacture the equilibrium the paper is supposed to test. We therefore place the stochastic-order verdict before the coefficient comparison and let it determine whether a long-run vector exists.

2 Data-First Inspection and Chilean Macro-Era Periodization

The canonical annual panel contains 105 observations for each capital component from 1920 through 2024. It joins the Chile capital-stock reconstruction to the official WBOP Chile series for trade, income, transfers, the current account, and foreign wealth. The WBOP workbook is frozen under SHA–256 `7198FE3F173056A849A5FD8F89F6FD99F598508D2B59C4749AF9A50E9BC91C7B`; its four principal accounting identities close to machine precision. Manufactured imports remain a broad import-dependence indicator, not a capital-goods series.

Figure 1: Chile Capital Stock Components in Log Levels with LOESS Inflection Overlays, 1920–2024

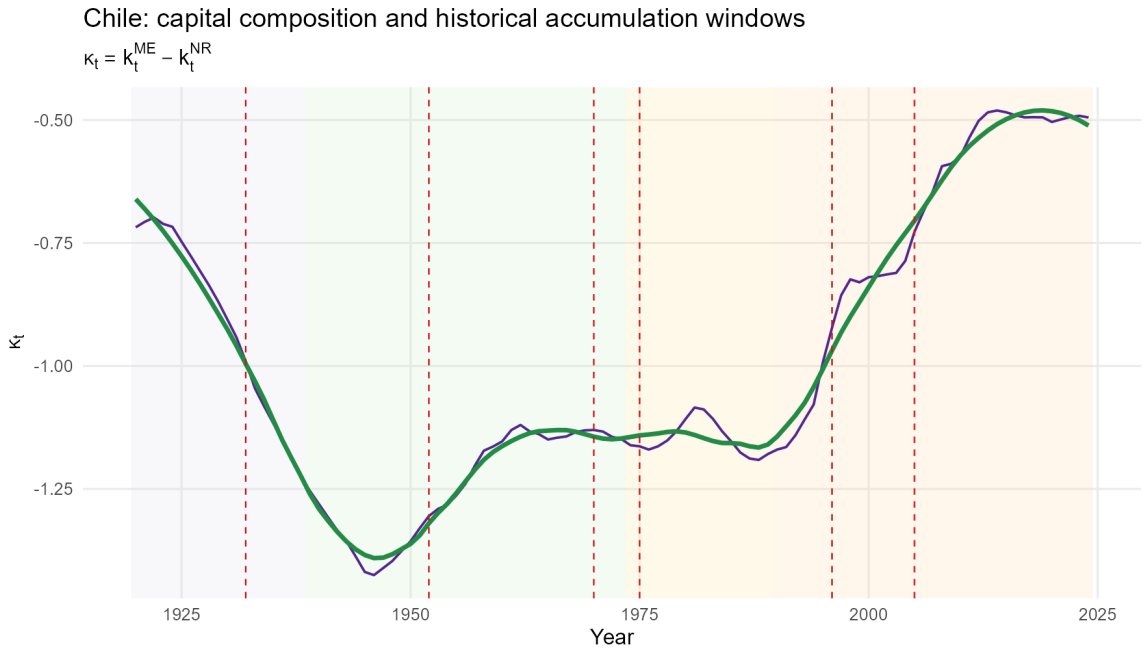


Note: Navy denotes machinery and equipment k_t^{ME}; crimson denotes nonresidential structures k_t^{NR}. Dashed lines are zero-crossings of the second derivative of the LOESS-smoothed capital-composition path. Source: canonical Chile capital panel.

Figure 1 shows parallel secular expansion without establishing a common stochastic trend. The two log stocks are smooth because they cumulate investment and retirement flows; this smoothness is precisely why their first differences require direct testing. The widening and subsequent narrowing between the paths appears more clearly in κ_t.

The LOESS second derivative turns in 1932, 1952, 1970, 1975, 1996, and 2005. These dates identify changes in the velocity of capital composition; they do not independently justify step dummies. The four historical windows therefore remain institutionally fixed, while formal break tests determine whether their internal parameters are stable.

Figure 2: Chile Capital Composition with LOESS Trend and Historical Accumulation Windows



Note: The purple line is $\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NR}}$; green is its LOESS trend. Shading denotes the Nitrate Crisis and Early State (1920–1938), Classic ISI (1939–1973), Dictatorship Shock (1974–1989), and Democratic Neoliberal Export Model (1990–2024). Red dashed lines are diagnostic curvature inflections, not imposed regime breaks.

3 Integration Orders and Cointegrating-Coefficient Admissibility

Each capital series is tested in levels, first differences, and second differences using ADF, Phillips–Perron, KPSS, ERS–GLS, and Zivot–Andrews procedures under constant and trend specifications. A transformation passes only when at least six valid tests are available and at least 60 percent support stationarity. An $I(1)$ verdict requires nonstationary levels and stationary first differences; $I(2)$-risk requires nonstationary first differences and stationary second differences. Conflicting short-window evidence remains unresolved.

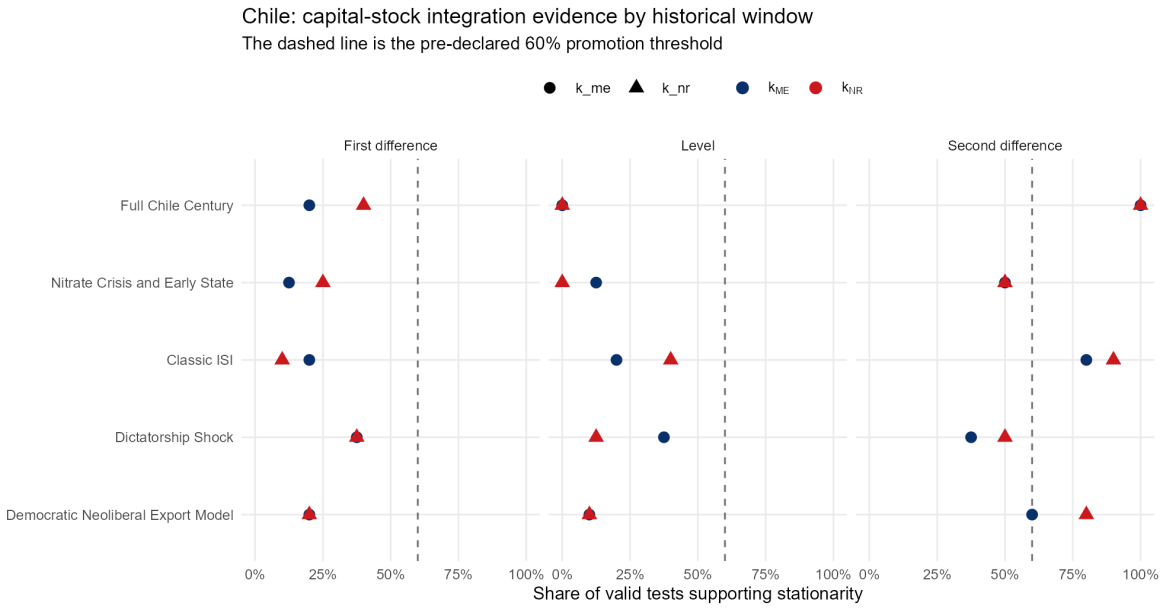
Table 1: Chile Capital Integration-Order and Standard-Model Admissibility Matrix

Historical Window	**Years**	**N**	k^{ME}	k^{NR}	**Standard Model**
Full Chile Century	1920–2024	105	$I(2)$-risk	$I(2)$-risk	**Blocked**
Nitrate Crisis and Early State	1920–1938	19	Unresolved	Unresolved	**Blocked**
Classic ISI	1939–1973	35	$I(2)$-risk	$I(2)$-risk	**Blocked**
Dictatorship Shock	1974–1989	16	Unresolved	Unresolved	**Blocked**
Democratic Neoliberal Export Model	1990–2024	35	$I(2)$-risk	$I(2)$-risk	**Blocked**

Note: A blocked window fails the common-$I(1)$ prerequisite for standard level cointegration. The two short windows would require 999-replication residual bootstrap inference if they became admissible; no model exists to bootstrap after the integration screen. Full statistics, controls, lags, and critical values remain in the quantitative ledger.

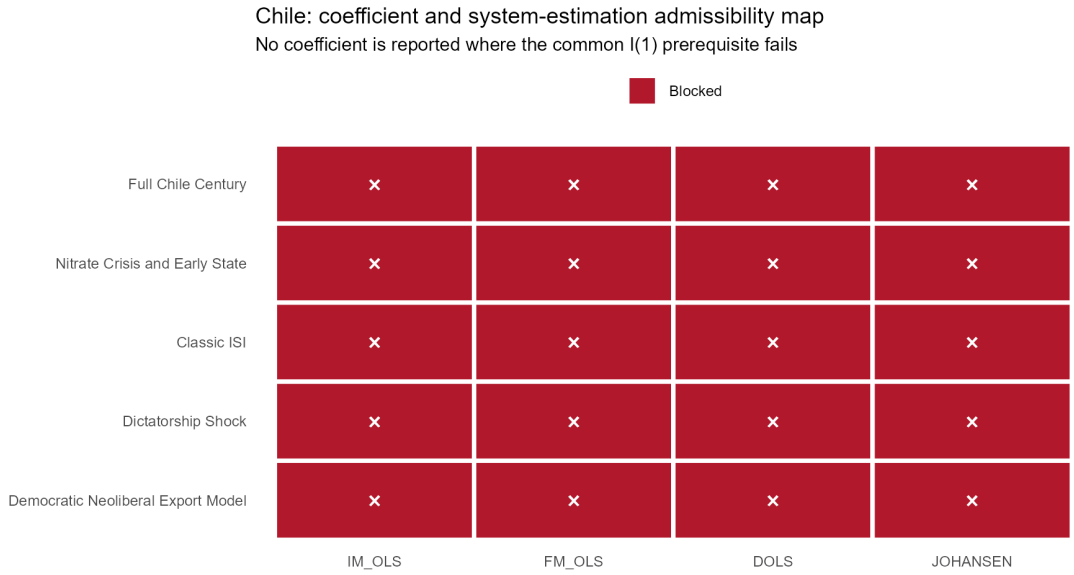
Table 1 and Figure 3 overturn the premise needed for a Chilean replica of the United States coefficient matrix. In the full century, machinery receives 2 of 10 stationarity votes after first differencing and structures receive 4 of 10; both receive 10 of 10 after second differencing. Classic ISI and the post-1990 model show the same ordering. The result is $I(2)$-risk, not weak $I(1)$. The crisis windows are too short and internally unstable to resolve the order. No historical window can promote β_{ME}.

Figure 3: Capital-Stock Stationarity Evidence across Transformations and Historical Windows



Note: Points report the share of valid tests supporting stationarity. The dashed line is the pre-declared 60 percent threshold. The full, ISI, and post-1990 samples cross the threshold only after second differencing.

Figure 4: Coefficient and System-Estimation Admissibility Map



Note: Red cells mark specifications blocked before estimation. IM–OLS is the intended baseline; FM–OLS and DOLS are diagnostic comparators; Johansen is the system benchmark. None executes when the component stocks fail the common-$I(1)$ requirement.

4 System Dynamics and the Absence of Admissible Adjustment Speeds

In an admissible system,

$$\Delta X_t = \boldsymbol{\alpha}\boldsymbol{\beta}' X_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i \Delta X_{t-i} + \boldsymbol{\epsilon}_t, \qquad X_t = (k_t^{\mathrm{NR}}, k_t^{\mathrm{ME}})', \tag{3}$$

the loading vector $\boldsymbol{\alpha}$ measures how structures and machinery correct disequilibrium. That interpretation depends on a valid rank and integration order; an estimated loading from a misspecified $I(1)$ system is

not an institutional adjustment speed.

Table 2: Chile Cointegrating-Coefficient and VECM Results Disposition

Historical Window	N	IM–OLS	FM–OLS	DOLS	Johansen / α
Full Chile Century	105	Blocked	Blocked	Blocked	Blocked
Nitrate Crisis and Early State	19	Blocked	Blocked	Blocked	Blocked
Classic ISI	35	Blocked	Blocked	Blocked	Blocked
Dictatorship Shock	16	Blocked	Blocked	Blocked	Blocked
Democratic Neoliberal Export Model	35	Blocked	Blocked	Blocked	Blocked

Note: "Blocked" means no coefficient was generated. It does not denote an insignificant estimate. Consequently, there is no rank, normalized β_{ME}, unit-vector restriction, error-correction loading, weak-exogeneity test, or system-stability statistic to interpret.

The empty coefficient space is the system result. It withdraws the former Chile elasticity near 0.88 and prevents rank-zero or wrong-order vectors from being presented as long-run parameters. The data may support an $I(2)$ system, a break-adjusted accumulation-rate relation, or a differently reconstructed capital stock, but none is equivalent to the standard level VECM requested by the United States template.

5 Structural Breaks and WBOP External Pressure

The failure is not produced by one isolated crisis year. The full-sample relation k_t^{NR} on k_t^{ME} has Bai–Perron breaks in 1934, 1949, 1964, and 1994, while recursive CUSUM rejects stability. Within-window tests also reject stability, including a 1983 break inside the dictatorship window and several shifts inside Classic ISI and the post-1990 period.

Table 3: Capital Breaks, LOESS Inflections, and WBOP Financing Compression

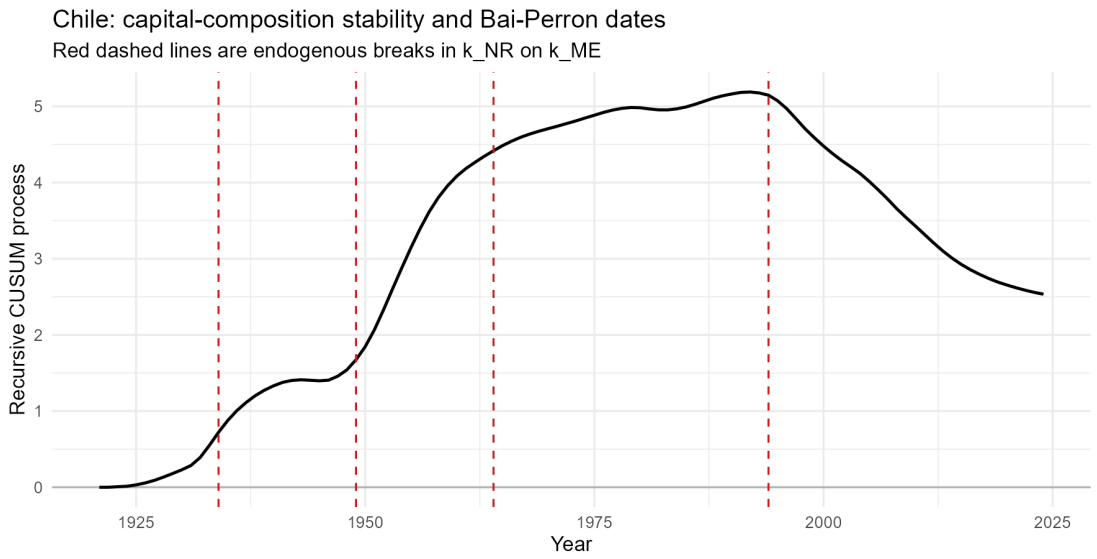
Evidence	Dates	Empirical Disposition
Bai–Perron capital relation	1934, 1949, 1964, 1994	Endogenous instability; no automatic step controls
LOESS curvature inflections	1932, 1952, 1970, 1975, 1996, 2005	Diagnostic periodization markers only
WBOP financing compression	1937, 1976, 1999, 2023	Outcome-independent events; never permanent regimes

Note: Persistent step or slope controls require formal and institutional concurrence. Nearby dates inform interpretation but are not treated as identical events.

WBOP enters only after this capital adjudication. Manufactured imports/GDP, the current account/GDP, and net foreign income/GDP pass as $I(1)$ and could enter one at a time in a valid long-run system. Gross foreign liabilities/GDP passes as $I(0)$ and is restricted to a short-run or impulse role. None can repair the failed capital order, so no WBOP interaction or trivariate vector is estimated.

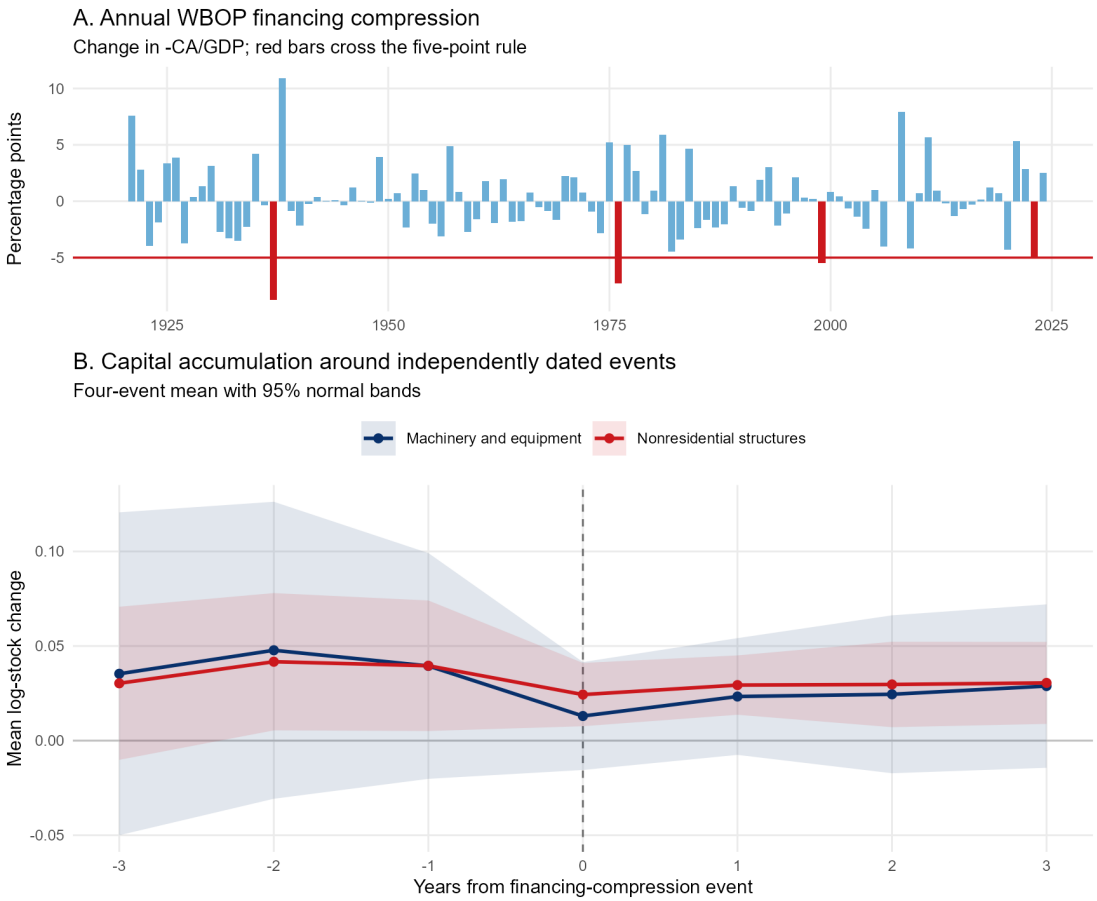
The event-year mean falls to 1.30 percent for machinery and 2.43 percent for structures, reducing $\Delta\kappa_t$ by 1.13 percentage points. Across event years and the following three years, the mean machinery-minus-structures gap is -0.57 percentage points. The 999-draw interval spans $[-2.36, 0.55]$, so the direction is consistent with machinery sensitivity but the four-event evidence does not identify a stable differential response. Threshold sensitivity strengthens the same caution: the event count moves from 14 at three points to two at six points.

Figure 5: OLS-CUSUM Path and Bai–Perron Break Dates in the Chile Capital Relation



Note: The black line is the recursive empirical fluctuation process from k_t^{NR} on k_t^{ME}. Red dashed lines mark endogenous Bai–Perron dates. The break evidence rejects an invariant full-century coefficient.

Figure 6: Annual WBOP Financing Compression and Capital Accumulation around Events



Note: Panel A dates annual falls of at least five GDP points in the external-financing proxy $-CA/Y$; these are not gross-flow sudden stops. Panel B reports mean Δk_t^{ME} and Δk_t^{NR} in $[-3, +3]$-year windows. Growth and capital outcomes never enter the event definition.

6 Admissibility and the A04 Measurement Boundary

The WBOP evidence sharpens A04 without supplying its latent conversion ratio. WBOP observes trade, income, transfers, the current account, and foreign-wealth positions. It does not jointly observe the historical machinery-import requirement, usable capital inflows, reserve use, debt service, and essential imports needed to construct F_t^{avail}, F_t^{req}, or ϕ_t. No exports-only envelope or import-intensity calibration is introduced.

The research establishes three bounded conclusions. First, Chile's capital composition is historically structured but does not satisfy the standard $I(1)$ conditions required for a level cointegrating coefficient. Second, formal breaks and WBOP events cluster near several changes in κ_t, but their dates are not interchangeable and do not define permanent regimes. Third, machinery accumulation weakens relative to structures around the four financing-compression events, yet the small-event interval includes zero. External pressure remains a plausible condition on the realization of mechanization; it is not an identified causal coefficient in this release.

Break adjustment sharpens rather than reverses this verdict. After removing one endogenous step and slope change, residual ADF tests reject a unit root in both capital levels for the full century, the Nitrate window, the dictatorship window, and the post-1990 window. That evidence points toward break-stationary levels, not a common $I(1)$ system. Classic ISI remains mixed: the adjusted structures level rejects a unit root while machinery does not. Because standard cointegration requires integrated levels rather than separately break-stationary or mixed-order series, the break-dummy evidence cannot be used to reopen Johansen or modified OLS.

Capacity utilization μ_t is not reconstructed here. Stationarity of κ_t would not by itself identify productive capacity, and the present evidence does not even establish that stationarity. The next admissible empirical move is a separately governed $I(2)$ or accumulation-rate design, not the restoration of a blocked $I(1)$ regression.

Appendix: Correct Normalization, Drift Alignment, and the Unit Restriction

Let the deterministic components of the two log stocks be

$$k_t^{\mathrm{NR}} = g_{\mathrm{NR}} t + u_t^{\mathrm{NR}}, \qquad k_t^{\mathrm{ME}} = g_{\mathrm{ME}} t + u_t^{\mathrm{ME}}. \tag{4}$$

Under the stated normalization, the expected cointegrating residual is

$$E[\varepsilon_t] = (g_{\mathrm{NR}} - \beta_{\mathrm{ME}} g_{\mathrm{ME}})\, t. \tag{5}$$

Eliminating deterministic drift therefore implies

$$\beta_{\mathrm{ME}} = \frac{g_{\mathrm{NR}}}{g_{\mathrm{ME}}}, \tag{6}$$

not its inverse. If machinery grows faster than structures, $g_{\mathrm{ME}} > g_{\mathrm{NR}}$, then $\beta_{\mathrm{ME}} < 1$. This is the interpretation maintained throughout the report.

Moreover,

$$\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NR}} = -\alpha + (1 - \beta_{\mathrm{ME}})\, k_t^{\mathrm{ME}} - \varepsilon_t. \tag{7}$$

Even if a rank-one system existed, $\kappa_t \sim I(0)$ would require the exact restriction $\beta_{\mathrm{ME}} = 1$. The redevelopment includes a likelihood-ratio restriction test in the estimator code, but it is not executed because no window reaches an admissible rank-one system.

References

Johansen, S. (1995). *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press.

Nievas, G., and Piketty, T. (2025). "Unequal Exchange and North–South Relations: Evidence from Global Trade Flows and the World Balance of Payments 1800–2025." World Inequality Lab Working Paper 2025/11.

Phillips, P. C. B., and Hansen, B. E. (1990). "Statistical Inference in Instrumental Variables Regression with $I(1)$ Processes." *Review of Economic Studies*, 57(1), 99–125.

Stock, J. H., and Watson, M. W. (1993). "A Simple Estimator of Cointegrating Vectors in Higher Order Integrated Systems." *Econometrica*, 61(4), 783–820.

Vogelsang, T. J., and Wagner, M. (2014). "Integrated Modified OLS Estimation and Fixed-b Inference for Cointegrating Regressions." *Journal of Econometrics*, 178, 741–760.